

EXECUTIVE SUMMARY

OVERALL RISK LEVEL

HIGH

THREAT CLASSIFICATION

ZERO_DAY_ANOMALY

PREDICTION CONFIDENCE

99.04374361042316

Intrusion Detection Event Analysis Report

Attack Type: ZERO_DAY_ANOMALY

Severity: HIGH

Risk Score: 84.19

Confidence: 99.04%

1. Executive Summary

A high-severity **Zero-Day Anomaly** has been detected with **99.04% confidence**, indicating a likely novel or previously unknown attack. The risk score of **84.19** suggests a significant threat to organizational assets. Immediate investigation and containment are required to mitigate potential damage.

2. Threat Analysis

- Nature of Attack:** Zero-day anomalies typically exploit unknown vulnerabilities, making them difficult to detect using traditional signatures.
- Indicators of Compromise (IOCs):**
 - Unusual process execution patterns
 - Anomalous network traffic (e.g., unexpected outbound connections)
 - Suspicious privilege escalation attempts
- Possible Attack Vectors:**
 - Exploit kits targeting unpatched software
 - Spear phishing with malicious payloads
 - Supply chain compromise

3. Potential Impact

- Operational:** Disruption of critical services, data exfiltration, or ransomware deployment.
- Financial:** Regulatory fines, incident response costs, and reputational damage.
- Security:** Lateral movement leading to further compromise of internal systems.

4. Recommended Actions

Immediate (Containment & Investigation)

- Isolate** affected systems to prevent lateral movement.
- Capture forensic artifacts** (memory dumps, logs, network traffic) for analysis.
- Engage threat intelligence teams** to identify possible threat actors or campaigns.

Short-Term (Mitigation & Hardening)

- Apply compensating controls** (e.g., network segmentation, stricter EDR rules).
- Review and update** intrusion detection signatures to flag similar anomalies.
- Conduct vulnerability scanning** to identify unpatched systems.

Long-Term (Prevention & Resilience)

- Enhance behavioral monitoring (UEBA, anomaly detection).
- Implement patch management for rapid zero-day mitigation.
- Conduct red team exercises to test defenses against novel attack methods.

Next Steps:

- Escalate to the incident response team for deeper analysis.
- Notify leadership and stakeholders per incident response protocols.

Prepared by: [Your Name]

Role: Senior SOC Analyst

Date: [Current Date]

This report is based on available telemetry and may require updates as new information emerges.

INCIDENT INFORMATION

SOURCE IP	DEST IP	PROTOCOL	DETECTION TIME
192.168.56.1	239.255.255.250	17	2026-07-07T00:46:28.371054

AI EXPLAINABILITY (SHAP)

TOP CONTRIBUTING FEATURES		
std_packet_size		+1.2243
duration		-0.6927
packet_count		-0.5299
mean_iat		-0.3450
total_bytes		+0.2279
mean_packet_size		+0.0247

SYSTEM RESPONSE

Anomaly Detected
Sparse Autoencoder MSE: 8.098776448979318e-05
Classification Complete
LightGBM assigned label: ZERO_DAY_ANOMALY
Policy Evaluated
Severity HIGH processed.
Action Executed

DRIFT & ADAPTATION

CONCEPT DRIFT STATUS
STABLE
ADWIN DETECTION
No significant distribution shift detected in risk scores.
MODEL RETRAINING REQUIRED
False

TECHNICAL EVIDENCE

RISK SCORE	ANALYST OVERRIDE
	Pending

84.18718206885968 /
100